

# Prathamesh Kalamkar

## Backend & AI Infrastructure Engineer

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research and engineering portfolio

### PROFILE

Backend engineer with 3.5+ years across Java, Spring Boot, Python, REST APIs, production support, and AI workflow infrastructure. Builds reliable tool-execution systems with explicit authority, idempotency, crash recovery, validation, observability, and deterministic tests. Current research examines whether security oracles used to evaluate AI-generated code measure executable behaviour or superficial form.

### EXPERIENCE

**Independent Software Engineer — AI and Automation Systems** Jan 2025–present

- Built Python services and workflow engines for tool-calling agents with structured outputs, durable state, retries, validation, human-review gates, and evidence-rich audit records.
- Developed GitHub-integrated developer tooling and OpenAPI migration infrastructure with least-privilege access, signed provider intent, bounded source transforms, and fail-closed coverage reporting.

**Software Engineer — matriXploit Pvt. Ltd.** Jun 2022–Jul 2024

- Built and maintained backend services using Java, Spring Boot, Python, SQL, and REST APIs; delivered features, production fixes, authentication refactors, diagnostics, and regression checks.
- Worked with QA and support teams on root-cause analysis, code review, release readiness, and production follow-up across shared systems.

**Software Engineer Intern — PhysicsWallah** Oct–Dec 2024

Implemented Java and Spring Boot backend features, investigated API behaviour, and supported testing and release readiness for high-usage product workflows.

**Technical Support Engineer — matriXploit Pvt. Ltd.** Jan–Jun 2022

Investigated application failures through SQL, logs, reproducible cases, and structured escalation; maintained reusable troubleshooting evidence for engineering teams.

### SELECTED ENGINEERING AND RESEARCH

#### SODA security-oracle benchmark

Reproducible construct-validity audit: 96 execution-labelled variants plus 140 real CVE fix pairs from 65 projects; CI, provenance, frozen protocol, and public v0.1.0 artifact.

#### PatchPilot

Provider-to-customer API migration system built for YC's Self-Maintaining APIs request: atomically manages auditable customer consent, verifies pinned keys, generates bounded JS/TS/Python patches, and opens reviewable PRs. 127 tests.

#### Effect Broker

Durable correctness boundary for side-effecting agent tools using stable intent, idempotency contracts, authoritative reconciliation, crash recovery, and explicit unknown outcomes.

#### Agent Redteam

Authorised adversarial testing and runtime guardrails for LLM agents with planted-evidence oracles, recomputable risk vectors, causal proof records, and offline regression suites.

### EDUCATION

**MSc Cybersecurity, First Class Honours** — Dublin Business School 2024–2025

**BSc Computer Science, First Class** — Savitribai Phule Pune University 2021–2024

### TECHNICAL SKILLS

|             |                                                                                               |
|-------------|-----------------------------------------------------------------------------------------------|
| Backend     | Python, Java, Spring Boot, FastAPI, REST APIs, SQL, PostgreSQL, SQLite                        |
| AI systems  | tool calling, MCP, agent workflows, structured outputs, RAG, evaluation harnesses, guardrails |
| Reliability | idempotency, durable execution, event sourcing, retries, audit trails, observability, pytest  |
| Platform    | GitHub Actions, Docker, Kubernetes, AWS, Azure, Terraform, CI/CD                              |

Project evidence, limitations, reproducible commands, and source repositories are linked from the portfolio. Research artifacts are not represented as peer-reviewed publications.